Group 2

Sales Analysis on Retail.csv

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1. Understanding the dataset

Retail.csv

The dataset contains detailed information on retail transactions, encompassing customer data, product details, order specifics, and shipping-related information.

It consist of 53 columns and 18,0519 entries.

1. Understanding the dataset

Float64	int64	object	
• Benefit per order • Sales per customer • Latitude • Longitude • Order Item Discount • Order Item Discount Rate • Order Item Product Price • Order Item Profit Ratio • Sales • Order Item Total • Order Profit Per Order • Order Zipcode • Product Description • Product Price	• Days for shipping (real) • Days for shipment (schedules) • Late_delivery_risk • Category ID • Customer Id • Department Id • Order Id • Order Customer Id • Order Item Cardprod Id • Order Item Id • Order Item Quantity • Product Card Id • Product Category Id • Product Status	• Type • Delivery Status • Category name • Customer City • Customer Country • Customer Email • Customer Fname • Customer Lname • Customer Password • Customer Segment • Customer State • Customer Street • Department Name • Market • Order City • Order Country	• Order date (DateOrders) • Order Region • Order State • Order Status • Product Image • Product Name • Shipping date (DateOrders) • Shipping date (DateOrders) • Shipping Mode

2. Exploratory Data Analysis (EDA)

- Distribution of Shipping Mode
- Distribution of sales across market
- Distribution of Product and Shipping Mode
- Monthly Sales Trend across the year

2. Exploratory Data Analysis (EDA)

Logistics Insight

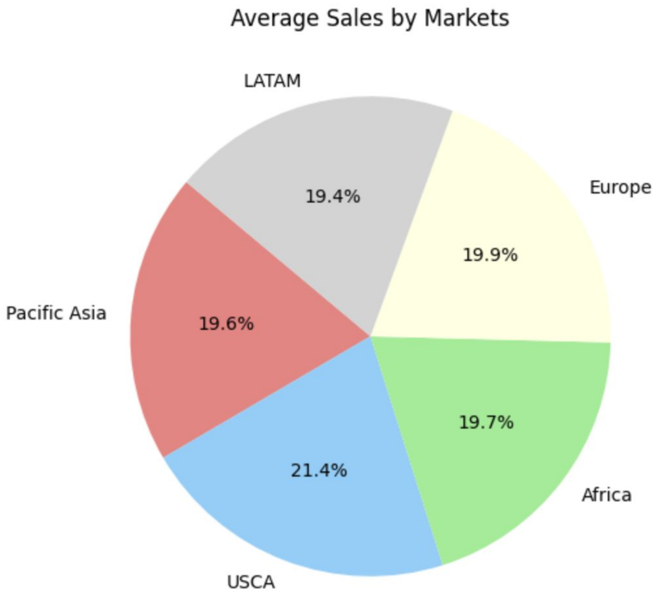
The most common shipping mode is Standard Class with 107,752 orders. This is followed by Second Class (35,216), First Class (27,814) and Same Day (9,737).

Preference for shipping mode can be influenced by cost, availability etc.

2. Exploratory Data Analysis (EDA)

Market Performance

USCA region leads in highest average sales contribution with a significant 21.4%, followed by Europe, Africa, and Asia-Pacific region.

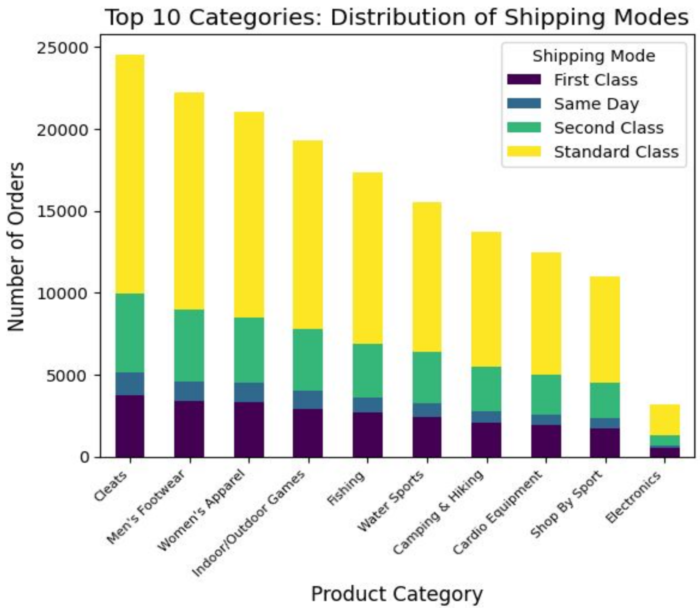


2. Exploratory Data Analysis (EDA)

Distribution of Product and Shipping Mode

Cleats emerge as the top selling product category for the retail business, followed by men’s footwear

Standard class shipping mode is the most common utilized option across all the product categories

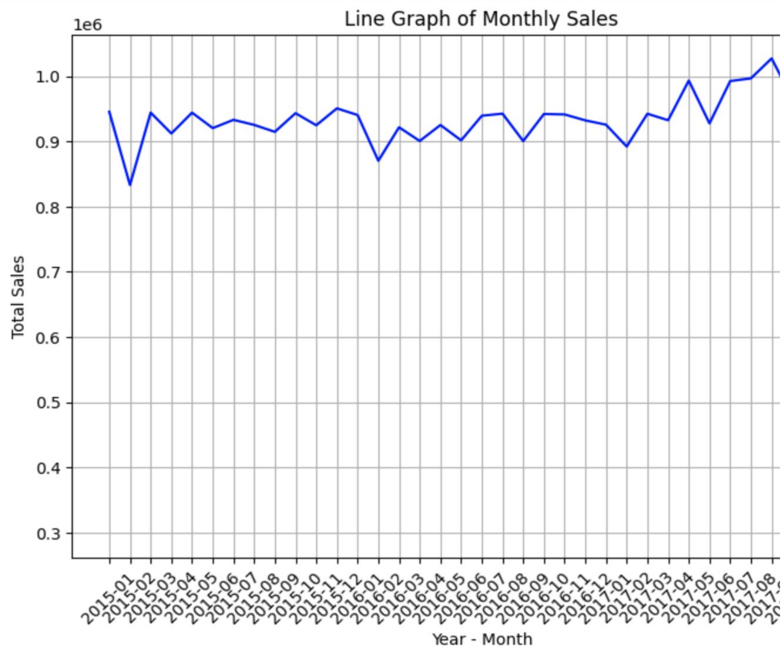


2. Exploratory Data Analysis (EDA)

Monthly Sales Across the year

The line graph shows the sales trend across the year from 2015 to 2017

There will be a significant drop in sales from december to january each year.



3. Problem Statement

- Predict the future sales for a retail business

Overarching:

- Build a predictive model to estimate sales for future orders based on historical sales data and customer information

4. Data Preparation

Data Preprocessing

- Check for **null value**
  - “Order Zipcode” & “Product Description” contains null values
- Drop **irrelevant** columns
- Use Pipeline to convert **categorical variables to numerical variables**
  - “Category Name” & “Order Region” are categorical variables

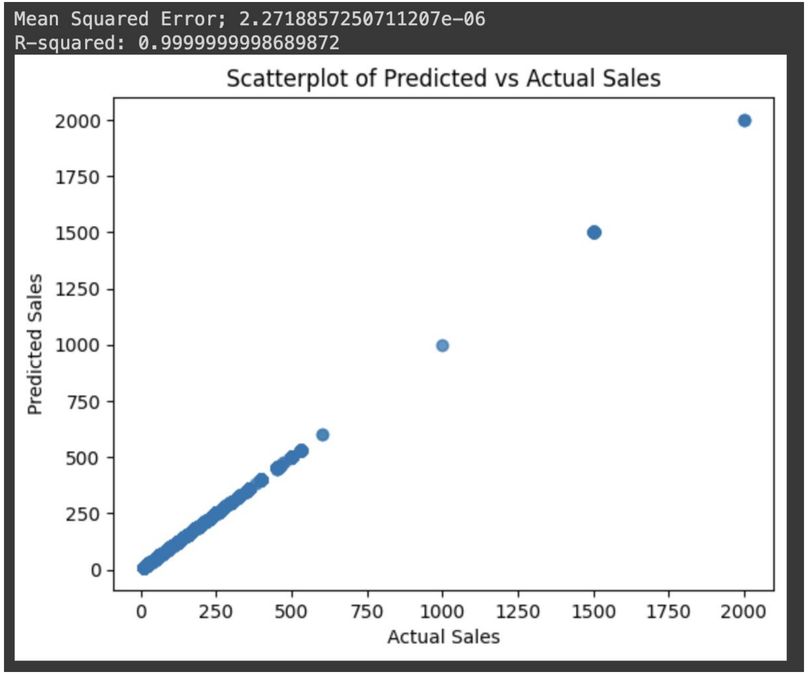
4. Data Preparation

Independent Features	Target Feature
• Sales per Customer • Order Item Quantity • Order Item Product Price • Order Profit Ratio • Order region • Category name • Product Price • Order Profit per Order • Order Item Discount	• Sales

5. Data Discovery

Problem

- Resulting scatterplot **did not** provide us with any conclusive insights
  - MSE: 0
  - R^2-Value : 1
- Model exhibiting signs of **multicollinearity**



5. Data Discovery

Multicollinearity

Definition: occurrence of **high intercorrelations** among **two or more independent variables** in a multiple regression model

Problem: affects **accuracy** of the model results.

- Difficulty in interpretation
- Inflates the standard errors

5. Data Discovery

Solution

1. Perform a check for **multicollinearity** using **Variance Inflation Factor (VIF)**
2. Perform **Hypothesis Testing** on our variables
3. Creation of an **Interaction Term** for 2 variables exhibiting high correlation
4. Perform another subsequent Variance Inflation Factor (VIF) check
5. **Train** the model using these revised features

5. Data Discovery

Variance Inflation Factor (VIF)

Definition: a mathematical formula that measures the **multicollinearity** among the **independent variables**

- VIF = 1 : **not** correlated
- 1 < VIF < 5 : **moderately** correlated
- VIF > 5 : **highly** correlated

	features	VIF
0	Sales per customer	12.030608
1	Order Item Quantity	6.607412
2	Order Item Product Price	inf
3	Order Item Profit Ratio	3.237290
4	Product Price	inf
5	Order Profit Per Order	3.296001
6	Order Item Discount	1.683199
7	Order Region	2.538609
8	Category Name	2.627952

5. Data Discovery

Ordinary Least Squares (OLS) Linear Regression

Definition: statistical method used for modeling the **relationship between a dependent variable and one or more independent variables**

	coef	std err	t	P>|t|
const	203.7719	1.15e-05	1.77e+07	0.000
Sales per customer	120.0438	1.41e-05	8.49e+06	0.000
Order Item Quantity	-0.0002	1.04e-05	-17.124	0.000
Order Item Product Price	-0.0001	8.41e-06	-15.365	0.000
Order Item Profit Ratio	1.274e-06	7.18e-06	0.177	0.859
Product Price	-0.0001	8.41e-06	-15.365	0.000
Order Profit Per Order	-4.669e-06	7.21e-06	-0.647	0.518
Order Item Discount	21.8004	5.2e-06	4.2e+06	0.000
Order Region	3.49e-07	5.73e-07	0.609	0.543
Category Name	6.393e-06	3.06e-07	20.857	0.000

5. Data Discovery

Observations

Variables with **high VIF**	**Not statistically significant** variables
• Sales per customer • Order Item Quantity • Order Item Product Price • Product Price	• Order Item Profit Ratio • Order Profit Per Order • Order Region

5. Data Discovery

Pearson Correlation

Definition: a measure of the **linear association** between **two continuous variables** (“Sales per customer” & “Product Price”)

Pearson correlation coefficient: 0.78178

	features	VIF
0	Sales per customer	11.702088
1	Order Item Quantity	6.547268
2	Product Price	17.231019
3	Order Item Discount	1.674327
4	Category Name	1.035500

5. Data Discovery

Interaction Term

Definition: a product term created by **multiplying two or more variables together**

- If two variables are **strongly correlated**, the product of these variables (interaction term) can introduce **new information** and **reduce the collinearity** between the original variables

	revised_features	VIF
0	Order Item Quantity	1.060556
1	Order Item Discount	1.246565
2	InteractionTerm	1.279176
3	Category Name	1.012698

6. Machine learning Model

6.1 Linear Regression Model

6.2 Classification Model

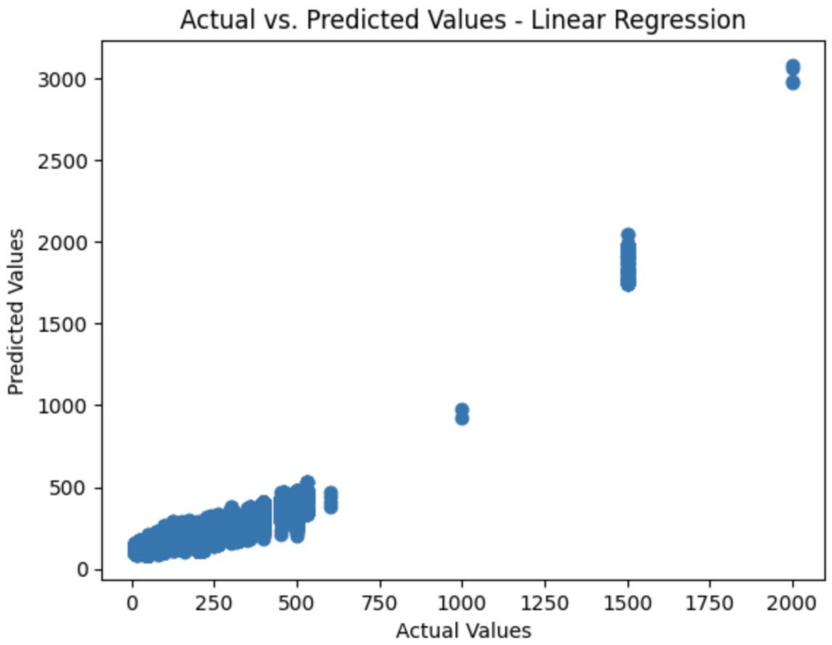
What did our results tell us about the accuracy of our trained model?

6.1 Linear Regression Model

Coefficient of Determination (R^2-Value)

Definition: a measure of how well **observed (test) outcomes** are replicated by the **(train) model**

```
The r2_score is :  0.7240438064123428
```



6.1 Linear Regression Model

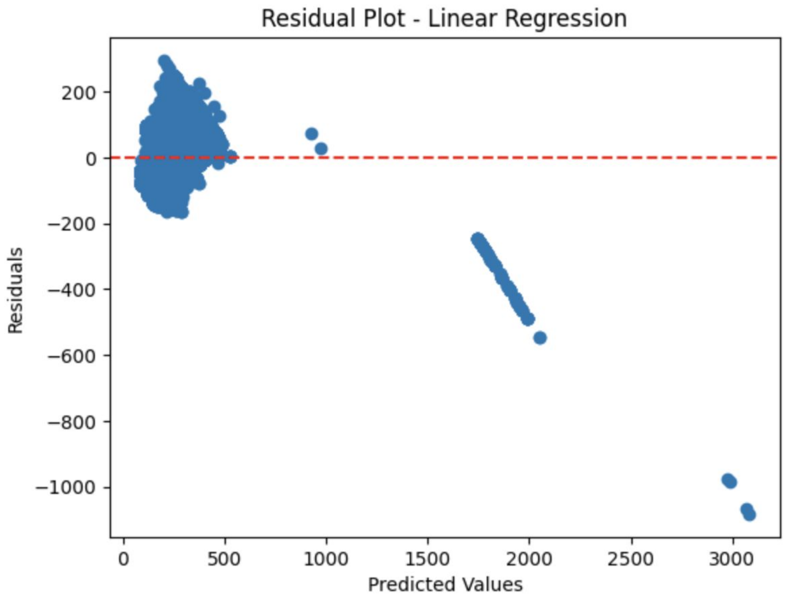
Residual Analysis

Definition: **differences** between the **observed values** (actual data points) and the **values predicted** by the model (fitted values)

Residual = Observed - Predicted

The distance from the line of 0 in the plot on the right indicates how bad the prediction was for that value.

```
The RMSE is :  69.20375721433847
```



Key interpretations from our Model

6.1.1 Key Interpretations

Our Insights

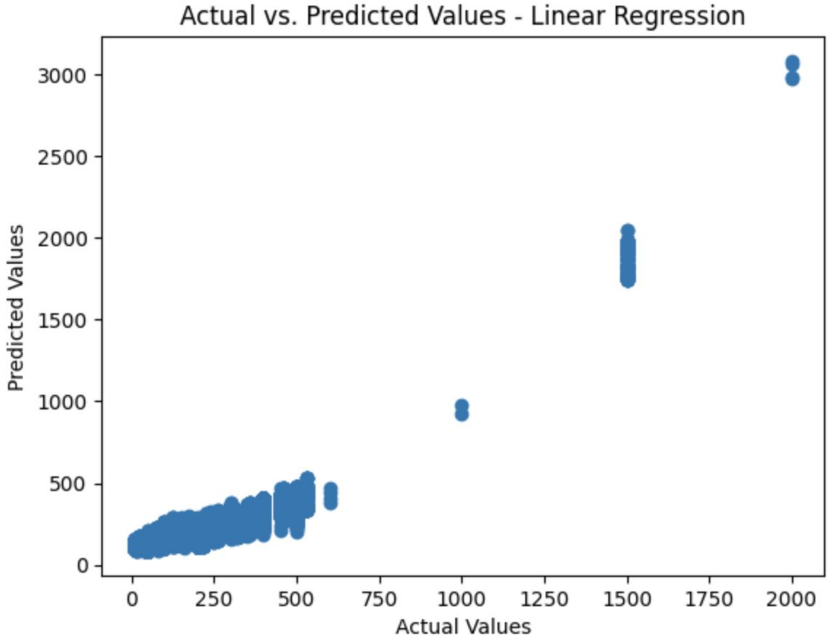
- **InteractionTerm** (Sales per Customer and Product Price) has the **highest** coefficient.
    - **highest** impact on Sales, ceteris paribus
- Category Name has a **negative** coefficient.
    - **negative** impact, ceteris paribus, and the strength of the impact depends on the type of good bought.

	coef
const	247.7721
Order Item Quantity	25.2608
Order Item Discount	40.8995
InteractionTerm	81.0082
Category Name	-1.6426

6.1.2 Potential Improvements

Note on our Model

- Before running OLS, it is common practice to check if the **model fulfills the assumptions of OLS**. However, we did not do this step.
- Our actual vs predicted values are **not exactly linearly related**.



6.1.2 Potential Improvements

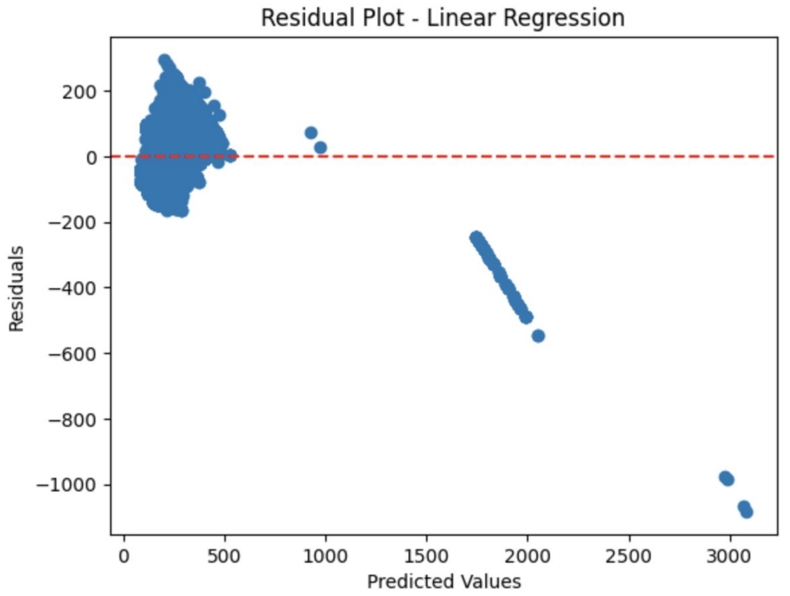
OLS Assumptions

1. **Linearity:** relationship between the independent and dependent variables
2. **Independence:** residuals are independent of one another
3. **Homoscedasticity:** residuals have a constant variance
4. **Normality of Residuals:** residuals are normally distributed
5. **No Perfect Multicollinearity:** independent variables are not perfectly correlated with each other

6.1.2 Potential Improvements

Note on our Model

- Residual plot is not randomly spread out. This can be explained through **Heteroskedasticity.**



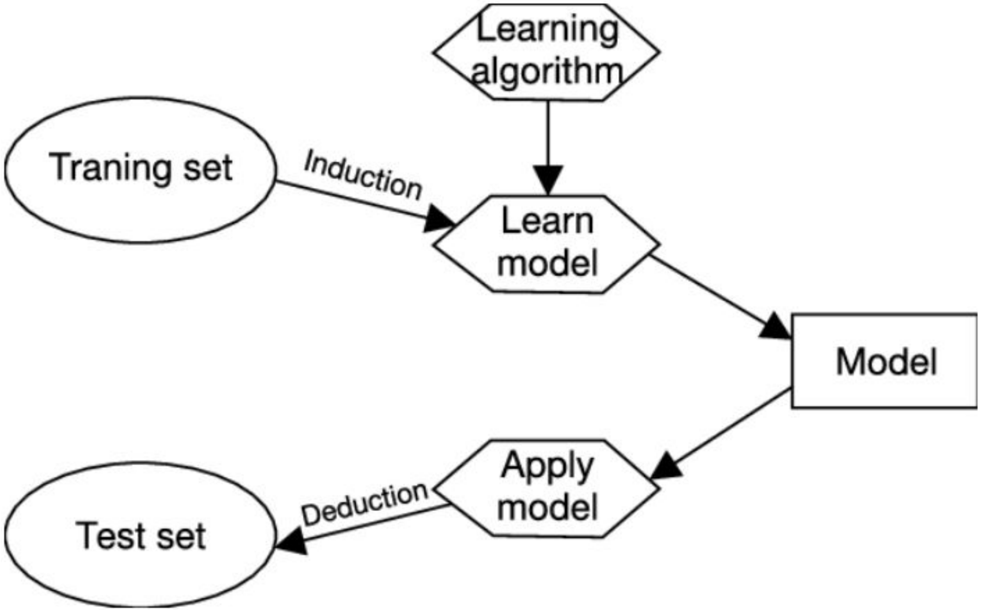
6.1.2 Potential Improvements

Accounting for Heteroskedasticity

- Definition: the **change in spread of residuals** over the **range of measured values**
- Exists is due to the **error variance changes proportionally** to a factor. Sometimes this factor might be a variable.
- Elimination: This can be done in a number of ways.
  - Using a measure to account for different population sizes of the data as they bring about unequal variance in the results
  - Why? Customers with higher purchasing power might spend more, but are fewer in number. Customers from less affluent countries might spend less, but are greater in number. Thus to account for this spread, we could potentially predict Sales Per Customer (or include a new variable such as Rate of Sales) instead of predicting Sales directly.

6.2 Classification Model

An alternative solution to maximising sales.



6.2 Classification Model

What is classification model?

- Classification is a supervised machine learning process, similar to regression, but is labelled as such when your target variable is **Discrete.**
- Classification comes in many forms:
  - Binary Classification - classify input data into 2 **distinct** categories; training data is input in a binary format (T/F, 1/0)
  - Multi-Class Classification - at least 2 **mutually exclusive** class labels, and we have to predict which class our input data belongs to.
  - Multi-Label Classification - predicting 2 or more classes for a given input data, note the labels within the given data are not mutually exclusive now (tldr : think multiple objects in a given input)

6.2 Classification Model

Our Initial Thought

- We wanted to swap Sales for **Category Name** as our target variable and see if we could classify future types of goods that were sold, based on the given features data.
- This was done with the belief that classification could be an alternative solution to **identify high performing products** based on trained features and capitalise on their popularity to boost sales.

6.2.1 Key interpretation

Decision Tree Classifier & Naive Bayes

- **Decision Tree Classifier** was the **most accurate** model, with K-Nearest Neighbours coming in second.
- Naive Bayes coming in last is potentially indicative of the relationship between our variables; Naive Bayes assumes conditional independence.

	Model	Score
3	Decision Tree Classifier	97.61
0	KNN	96.69
4	Support Vector Machines	81.05
2	Logistic Regression	73.30
1	Naive Bayes	59.62

6.2.1 Key interpretation

A couple of things we realized

- Our feature set might not necessarily have the best features to predict future Category Names as they were optimised to look at Sales.
- As an example, a type of good that historically generates high sales might not generate the same amount of high sales in the future (dependent on tastes and preferences) and therefore classifying based on sales may lead to erroneous results.

6.2.2 Potential Improvements

Inclusion of Other Features

- Instead of keeping the same features, we could have expanded our features/or used different ones in order to train a more accurate model
- A confusion matrix could also be plotted to understand where the model is making mistakes, distinguishing between true positives, false positives, true negatives, false negatives.
- We could also potentially use other measures of testing accuracy such as Precision, Recall and F1 Score.

Thank you!

From Group 2